

Total No. of Questions : 8]

SEAT No. :

PA-1451

[Total No. of Pages : 2

[5926]-67

T.E. (Computer Engineering)
ARTIFICIAL INTELLIGENCE
(2019 Pattern) (Semester - II) (310253)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Attempt Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Assume suitable data, if necessary.

Q1) a) Explain Min Max and Alpha Beta pruning algorithm for adversarial search with example. **[9]**

b) Define and explain Constraints satisfaction problem. **[9]**

OR

Q2) a) Explain with example graph coloring problem. **[9]**

b) How AI technique is used to solve tic-tac-toe problem. **[9]**

Q3) a) Explain Wumpus world environment giving its PEAS description. **[9]**

b) Explain different inference rules in FOL with suitable example. **[8]**

OR

Q4) a) Write an propositional logic for the statement, **[10]**

i) "All birds fly"

ii) "Every man respect his parents"

b) Differentiate between propositional logic and First order logic. **[7]**

P.T.O.

- Q5) a)** Explain Forward chaining algorithm with the help of example. **[9]**
b) Write and explain the steps of knowledge engineering process. **[9]**

OR

- Q6) a)** Explain Backward chaining algorithm with the help of example **[9]**
b) Write a short note on : **[9]**
i) Resolution and
ii) Unification

- Q7) a)** Write a short note on planning agent, state goal and action representation. **[6]**
b) Explain different components of planning system. **[6]**
c) Explain the components of AI. **[5]**

OR

- Q8) a)** What are the types of planning? Explain in detail. **[6]**
b) Explain Classical Planning and its advantages with example. **[6]**
c) Write note on hierarchical task network planning. **[5]**

